

Internet of Things

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Reference book



Chapter 7

Outline

- ▶ Introduction
- ▶ Sensor Technology
- ▶ Participatory Sensing, Industrial IoT and Automotive IoT
- ▶ Actuator
- ▶ Sensor Data Communication protocols
- ▶ Radio Frequency Identification Technology
- ▶ Wireless Sensor Networks Technology
- ▶ References

Objectives

- ▶ sensor technology for sensing the real world using analog and digital sensors, and examples for sensing devices for IoT and M2M
- ▶ Define the concepts of participatory sensing, industrial IoT and automotive IoT
- ▶ Describe the uses of actuators in devices
- ▶ Describe the uses of data communication using serial bus protocols
- ▶ Explain the Radio Frequency Identification (RFID) technology
- ▶ Explain the Wireless Sensor Network (WSN) technology

Sensor Technology

- ▶ RFIDs and their sensors play an active role in Internet of RFIDs. RFIDs are used for identifying objects using RF. Two examples of application of Internet of RFIDs are tracking parcels, goods and delivery, and supply chain management.
- ▶ A technology for designing the sensors and associated electronic readers, circuits, and devices.
- Data is used for analytics, visualisation, intelligence and knowledge discovery or controls and monitoring. Control systems use the sensors for monitoring and the actuators for actions.

Sensor Technology

It is a technology used for designing sensors and associated electronic readers, circuits and devices.

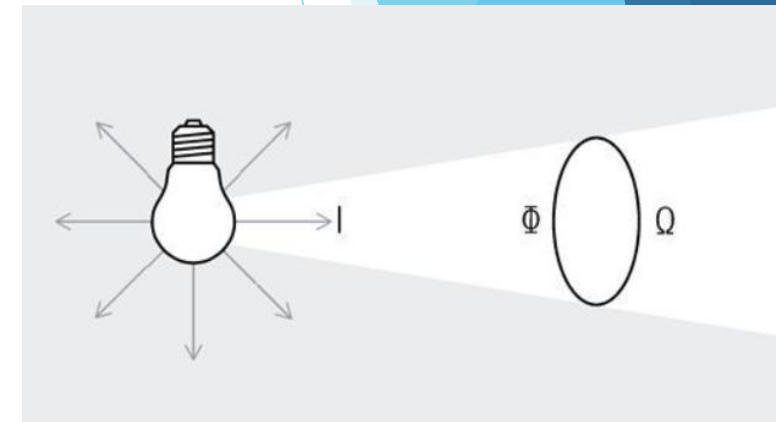
A sensor can sense a change in physical parameters, such as temperature, pressure, light, metal, smoke and proximity to an object. Sensors can also sense acceleration, orientation, location, vibrations or smell, organic vapours or gases.

A microphone senses the voice and changes in the sound, and is used to record voice or music.

A sensor converts physical energy like heat, sound, strain, pressure, vibrations and motion into electrical energy. An electronic circuit connects to the input at a sensor. The circuit receives the output of the sensor. The output is according to the variation in physical condition.

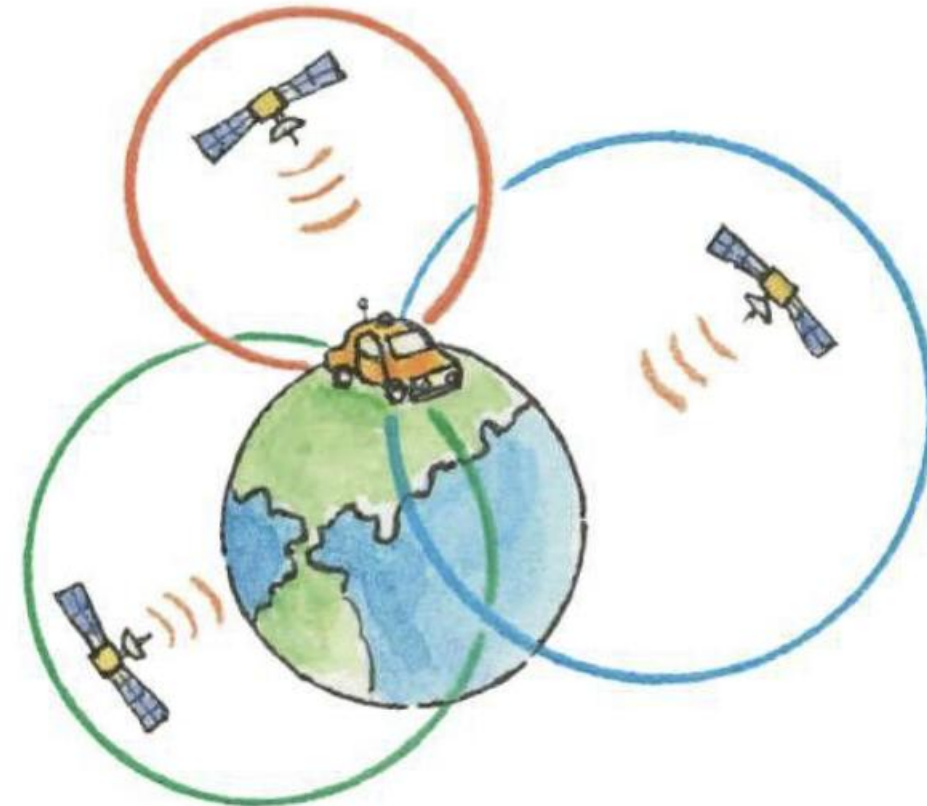
Physical parameter Sensing

- ▶ Senses a change in physical parameter
 - ❖ Temperature
 - ❖ Pressure
 - ❖ Light intensity
 - ❖ Metal proximity to an object (proximity sensor can detect metal targets)
 - ❖ Smoke in proximity
 - ❖ GPS



Physical parameter Sensing

- The circuit receives energy in form of variations through currents, voltages, phase angles or frequencies. Analog sensors measure the variations in the parameters with respect to a reference or normal condition and provide the value of sensed parameter after appropriate calculations.



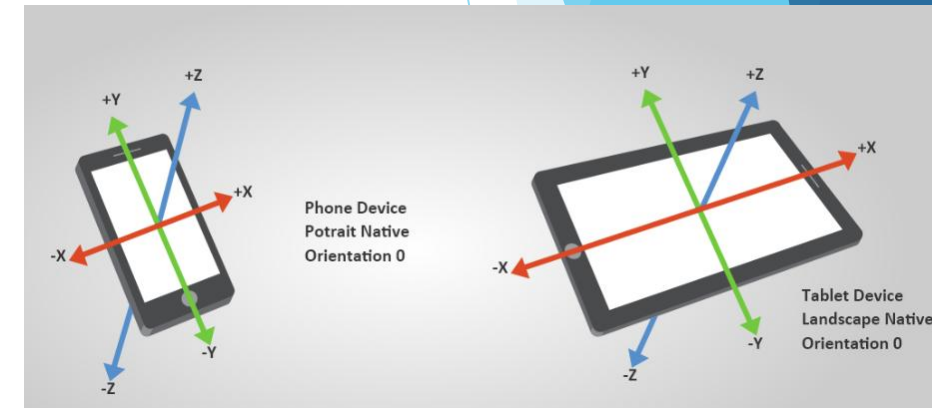
Physical parameter Sensing

A characteristic parameter of a circuit changes with the physical conditions. Technology that facilitates such changes due to sensing is also used in mobile phone. A mobile phone can sense surrounding conditions. The touchscreen of a mobile phone can sense a finger touch and gestures. Smartphones have resistive and capacitive sensors, photodiode current-based sensors, and acceleration, gyroscope, temperature and pressure sensors. The sensors enable the functioning of applications and games.

A microcontroller is an associated computing device with a sensor circuit which calculates the touched position and maps it to a user command when a resistive-based touchscreen is used.

Physical parameter Sensing

- ▶ Acceleration
 - Orientation
 - Location
 - Vibrations
 - Smell, organic vapours or gases.



Transducer as Sensor

- ▶ Converts physical energy like heat, sound, strain, pressure, vibrations and motion into the electrical energy
- ▶ Voice and changes in the sound, and is used to record voice or music
- ▶ Analog sensors generate analog outputs as per the physical environmental parameters such as temperature

The electronic circuit for the sensor

- ▶ Connecting to the input at a sensor
- ▶ Circuit input receives the output of a sensor/transducer
- ▶ The circuit output variation is according to the
- ▶ variation in physical condition

The output changes as per a change in the physical energy at input. An IC-based circuit for a temperature-transducer induces current in the output according to the heat energy, represented by the temperature.

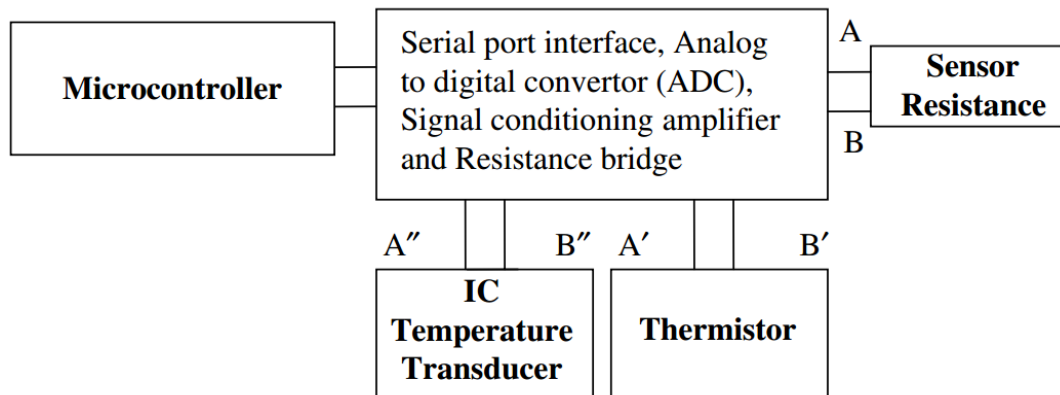


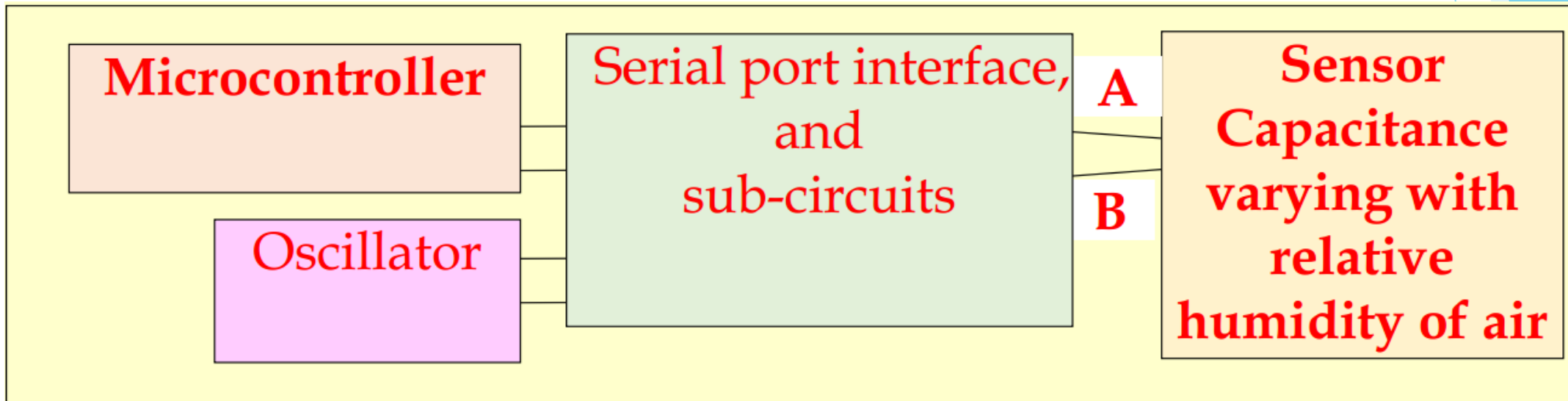
Fig. Microcontroller serial port connected to sub-circuits—serial port interface, ADC, signal conditioning amplifier, resistance bridge and sensor resistance outputs A and B. Alternatively circuit connects to a thermistor output at A and B

Sensor Circuit

- ▶ Receives the energy in form of variations in the currents, voltages, their phase-angles or frequencies.
- ▶ Sensor measures the variations in the parameters with respect to a reference or normal condition and provide the value of sensed parameter after appropriate calculations

#Analog sensor circuit connects to a signal conditioner amplifier, then to an analog to digital converter

Physical Parameter variation with the Physical condition at the sensor



The microcontroller electronic-circuit port connected to interface and sub-circuits; and receiving input from a capacitive sensor whose capacitance value is varying with the relative humidity of air and the circuit sensing the variation in relative humidity.

Smart Sensor

- ▶ A sensor with a built-in integrated circuit to directly give the output as physical parameter on giving input supply voltages and other signals as commands
- Digital sensor uses the sensor and an associated electronic digital circuit to read 1s and 0s
- Use of capacitance changes for measuring Relative Humidity % (relative percentage ratio)
- Use of time intervals between IR(inversion recovery) pulses and reflected pulses for measuring the distance of an object
- Use of photoconductor, p-n junction photodiode or phototransistor for measuring the intensity of light



Smart Sensors Examples

- ▶ A smart sensor for temperature giving output in term of hex-digit as 10 UART serial bits according to temperature in degree Celsius; 01100100 for 100 C and 00100100 for 36 C after taking calibration coefficients also into account.
- MEMS(Micro electromechanical systems) measures the vibrations and shocks by changes due to piezoelectric effect (Piezoelectric Effect is the ability of certain materials to generate an electric charge in response to applied mechanical stress)
- Pressure sensors use the resistance changes due to strain. Alternatively, they use piezoelectric effect

Smart Sensors

- ▶ Sensing the Things
 - Reading barcodes

A barcode is a representation of data. The data relates to the object where the printed code strip is attached. The code is read by an optical scanner.

- QR Code

QR code is an abbreviation for Quick Response Code. It was first used in automotive industry. Its applications are product identification, tracking, marketing and document management.



Smart Sensors

► Motion Sensors for Moving Objects

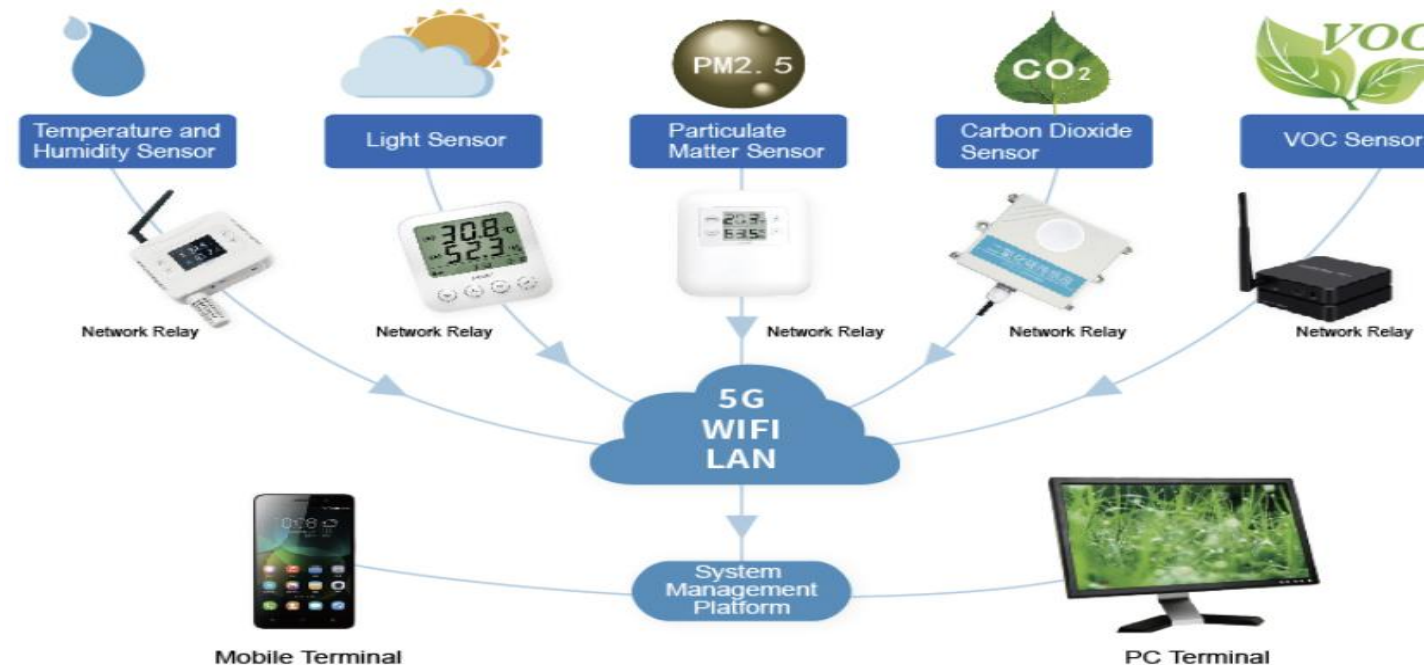
The sensor measures delay between successive reflected IR light pulses. An LED source is an IR light source and a phototransistor is an IR sensor. Alternatively, **ultrasonic wave echoes can be used to sense the motion of light. The sensor measures the delay between successive echoes.**



Smart Sensors

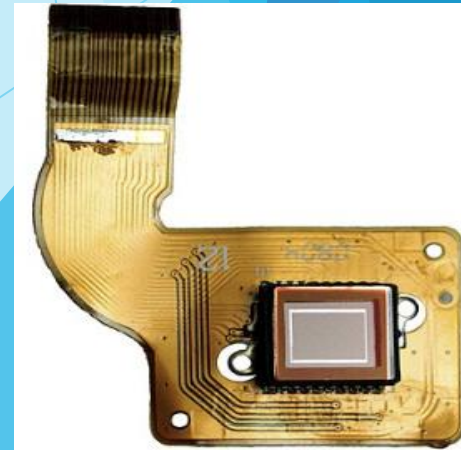
➤ Environmental Monitoring Sensor

Environment parameters are temperature, humidity, barometric pressure and light. A collective use of these parameter sensors enable monitoring of the environment. The data of these sensors adapts to the requirement and sends communication on the Internet to the cloud or web for the environment monitoring applications.



Smart Sensors

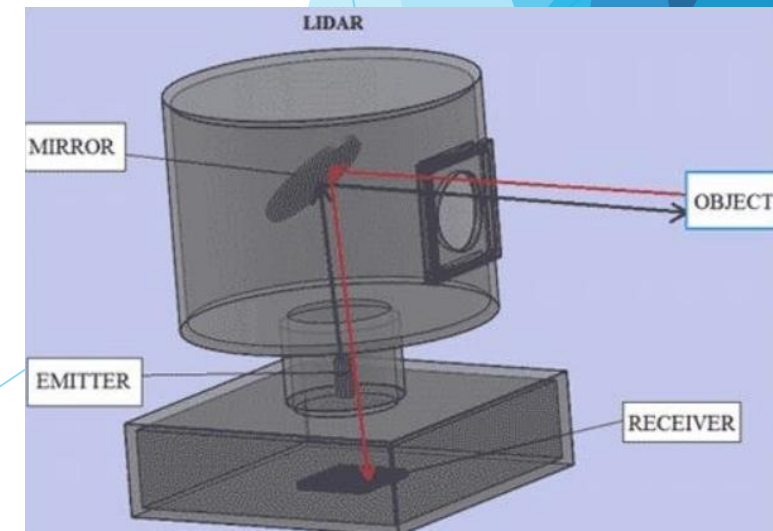
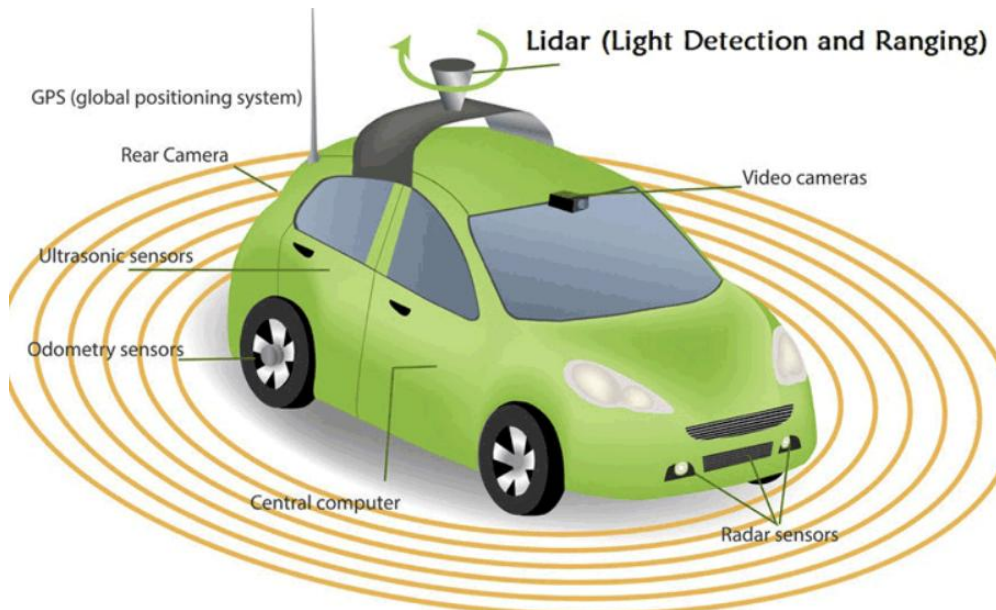
- ▶ Location Data
- ▶ GPS: Location determination can be done using a Global Positioning System (GPS), also known as Geographical Positioning System. A user can receive the location from a service provider. The service provider finds the GPS location through signals from satellites. The service provider finds the user location with respect to the service provider through its base stations.
- ▶ Camera: Camera is an image sensor. The camera uses CCD, which consists of a large number of pixels, exposed to the light from the image. It accumulates charges on each of the pixel present at a large number of horizontal and vertical coordinates. The charge accumulation is as per the intensity of light at the corresponding pixel coordinate in the image.



A CCD image sensor on a flexible circuit board

Smart Sensors

- LIDAR (Light + Radar) [Laser Imaging, Detection and Ranging] sensors and laser 3D imaging technology enables remote sensing and imaging. It finds distances by throwing light using laser on target. The sensor senses the reflected light which enables computations of distance.



Sensing process for IoT applications and services

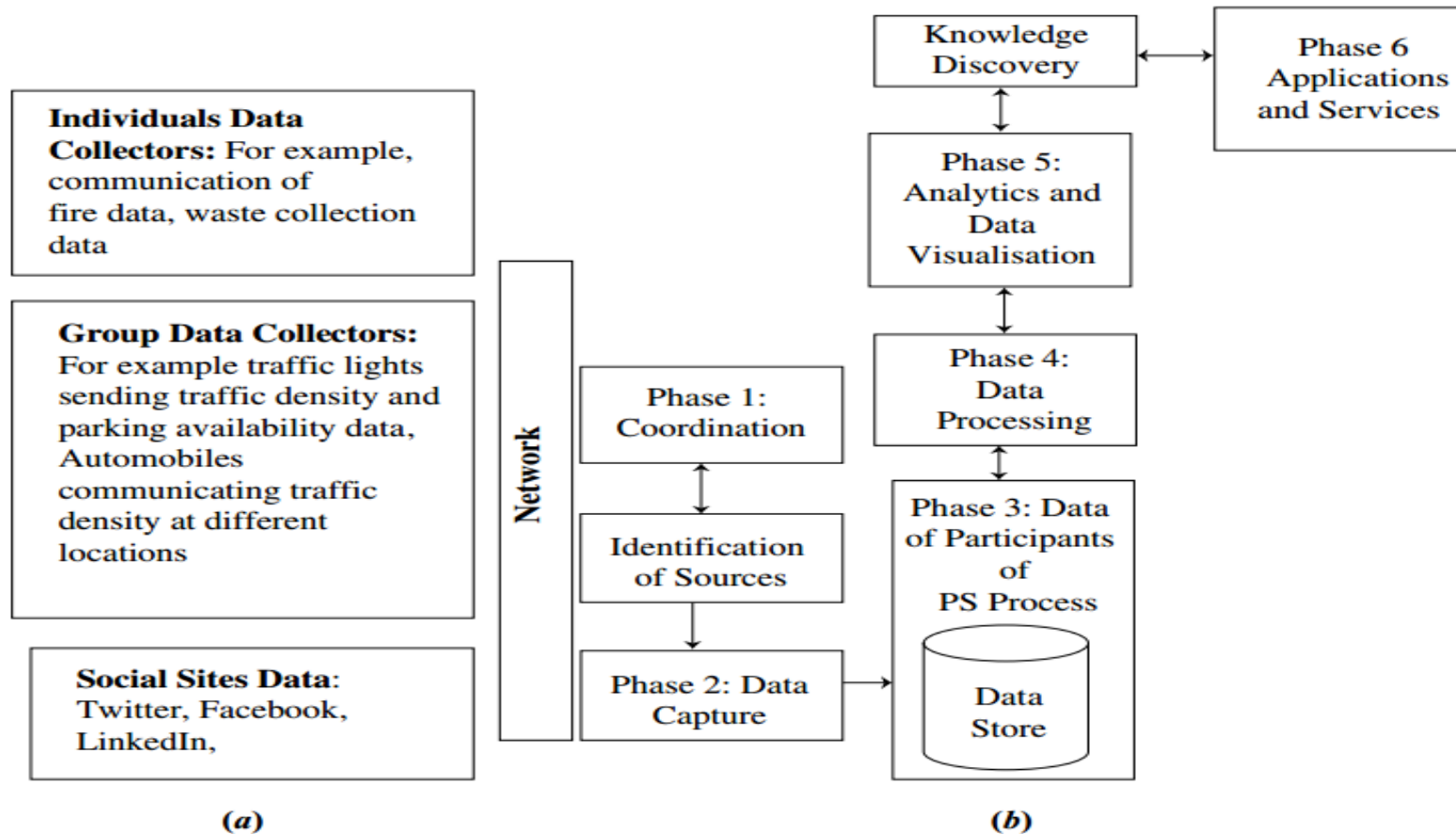


Figure 7.9 (a) Sources of data in the processes and (b) Phases of a participatory sensing process for IoT applications and services

IoT phases in the bicycle manufacturing process

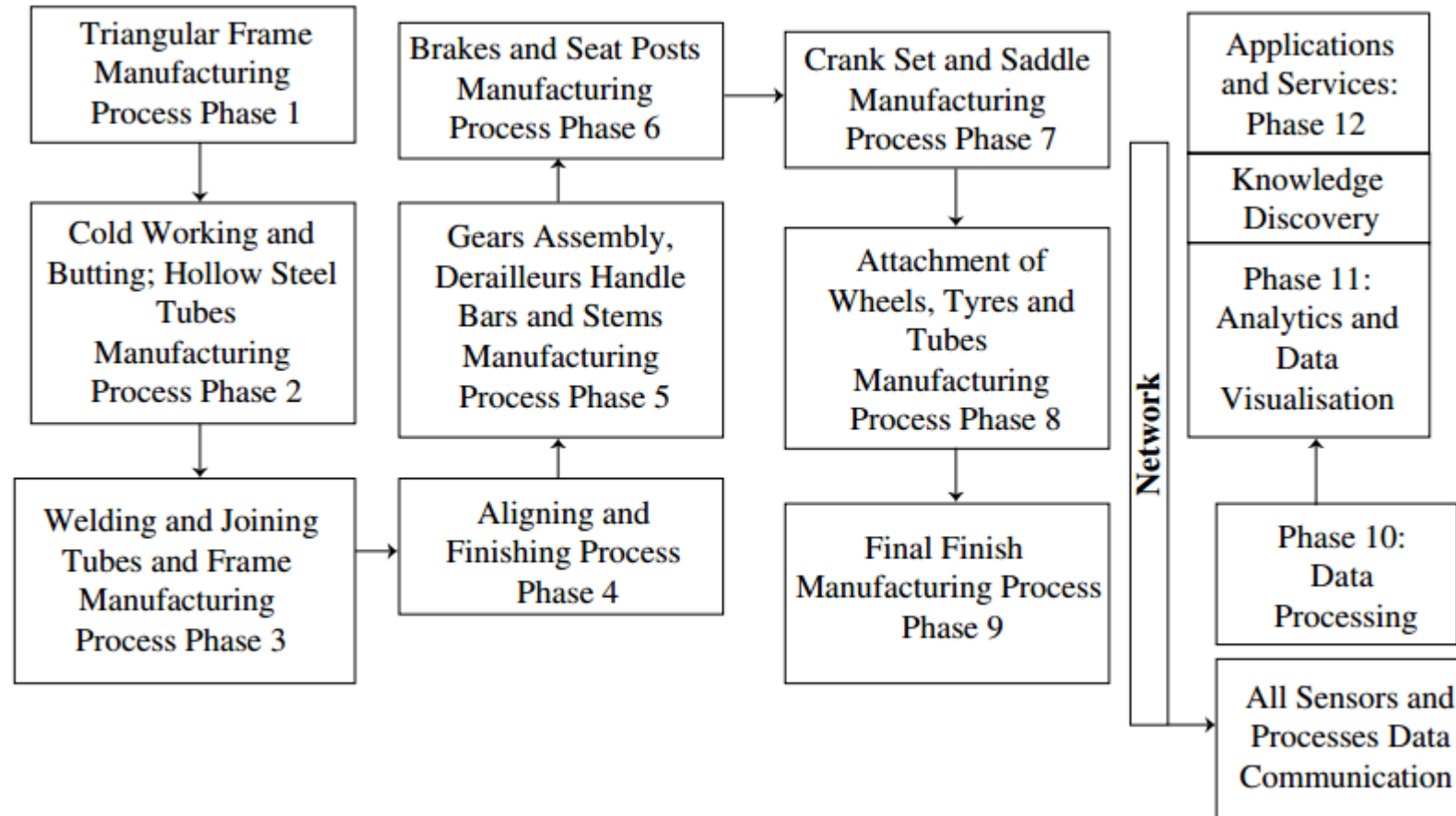


Figure 7.10 IoT phases in the bicycle manufacturing process

Sensing the change of a State

- ▶ Sensing circuit sensing 0 or 1 as per one of the two states. The change of state, for example,
 - ❖ light to darkness
 - ❖ finger touch
 - ❖ proximity or far
 - ❖ completion of a revolution of a wheel
 - ❖ tire pressure below threshold or above the threshold

Automotive IoT

Connected Cars Technology

- ▶ Display for driver that enables driving through the shortest route, avoiding the congested route, etc.
- ▶ Customization of functioning of the vehicle to meet the driver's needs and preferences
- ▶ Get notifications about traffic
- ▶ Protecting cars against theft
- ▶ Weather and enroute destinations
- ▶ Keeping a tab on driver's health and behavior.

Vehicle-to-Infrastructure Technology

- ▶ Alerts and warnings for forward collision
- ▶ Information about blind spots
- ▶ Notification about a vacant parking space
- ▶ Information about traffic congestion on route to destination
- ▶ Stream live music and news.

Automotive IoT

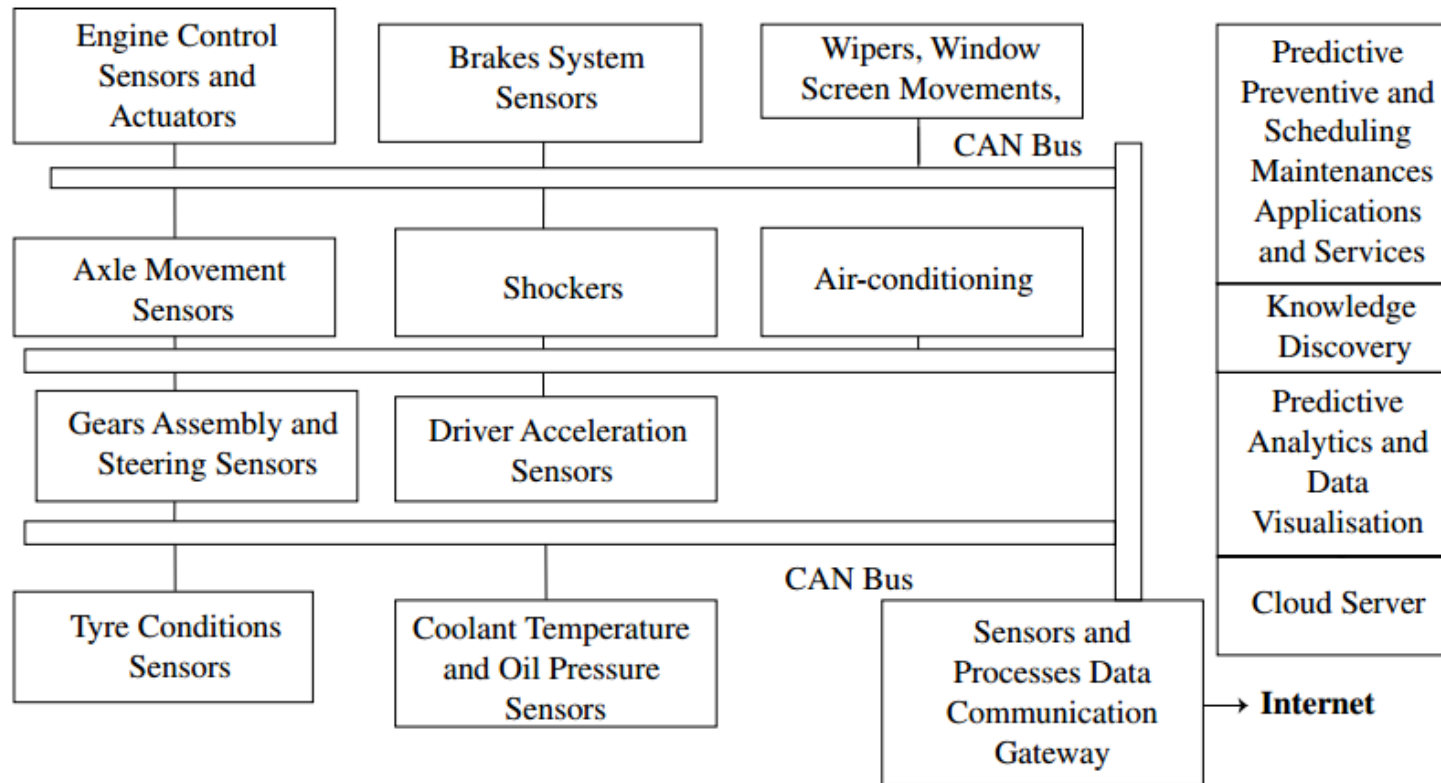


Figure 7.11 Internet of connected car components for predictive and preventive maintenances of automobile by service center

Summary

- ▶ We discussed about different Sensors
- Variation of a physical parameter with the physical condition to be sensed
- Sensing of temperature, pressure, vibrations, proximity, gas, relative humidity
- Sensing of circuit parameters: current, voltage, resistance, capacitance, phase angle or frequency
- Smart Sensor
- Integrated Circuit Multi-parameter Sensing Circuit Board

RFID - RADIO FREQUENCY IDENTIFICATION TECHNOLOGY

- An identification system using the tagging and labelling of objects
- RFID technology enables tagging with a product, parcel, postal article, person, bird, animal, vehicle or object
- Makes the identification feasible using RF
- An ID can use UART or NFC protocol, and identify the tag

RFID tag communication of ID

- ▶ ID communicate using UART or NFC protocol
- ▶ Identifies a tag, when at a distance less than 20 cm
- ▶ An active NFC device/mobile generates RF field and generates enough power for device RFID transmitter
- ▶ Using that power, the RFID transmits the identification tag contents.
- ▶ Active device has a built-in power source (battery) and transmits the information on its own.

Passive device

- ▶ Drives power from electrical current induced in its antenna by the incoming RF signals from a reader or hotspot
- ▶ Then the tag transmits information back.



Components needed in a system for the IoT Application and services

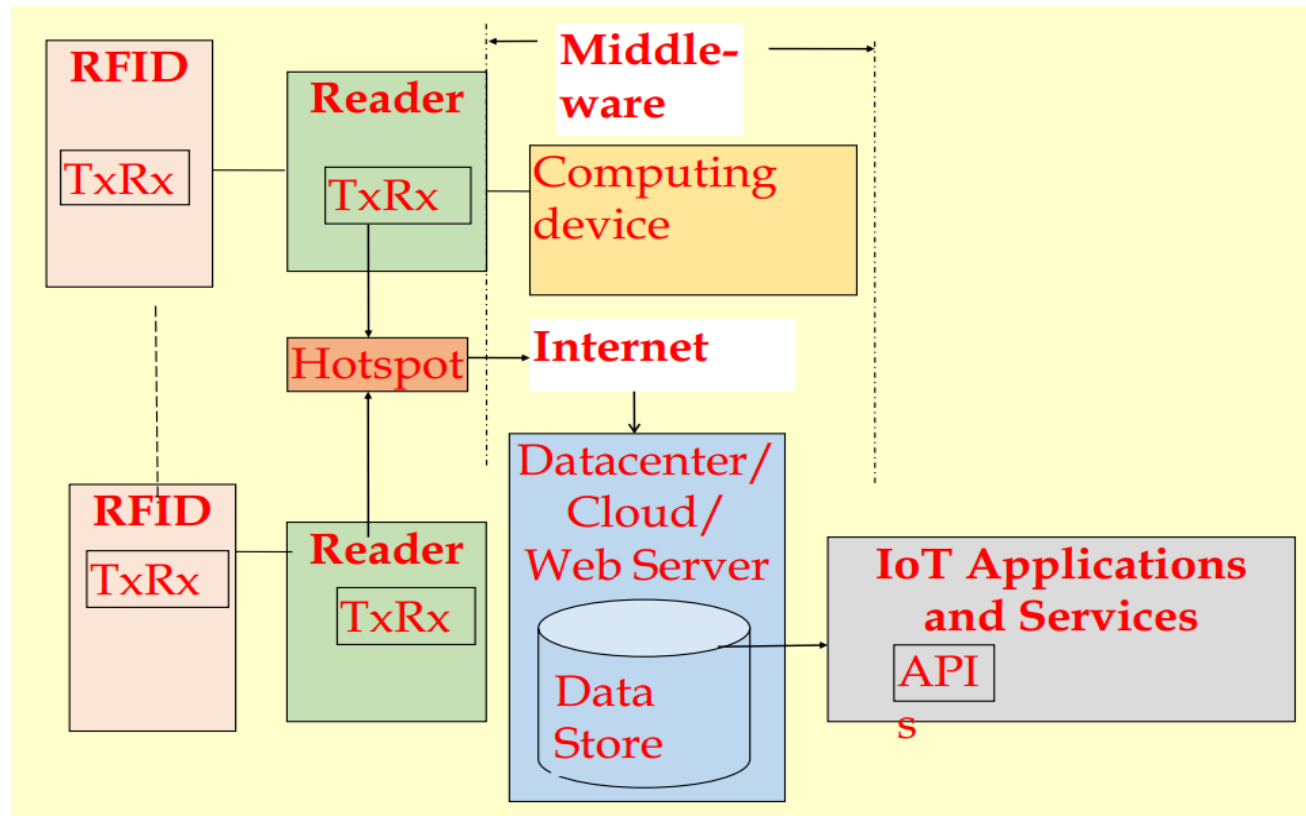


Fig. The Components needed in a system for the IoT Applications and Services; TxRx- Transmitter-cum-receiver

Middleware

- ▶ Software components used at the reader, read manager, data store for the transaction data store and APIs of the applications

Applications and services

- ▶ Other associated applications software use the Data Store at the cloud/web server

Design and Security Issues

- ▶ Design issue: Designing a unique ID system needs a standard global framework.
- ▶ Security issues: A tag is read only— Can thus interact with any reader and thus allows automated external monitoring

Privacy Issue

- ▶ A privacy issue arises when tag and reader need not to be authenticated before their use
- ▶ Full implementation of privacy and security needs data processing at the tag and reader with access encryption and authentication algorithms
- ▶ RFID system vulnerability to external virus attacks.

Cost and Protection

- ▶ RFID tag and reader become costly with data processing and security
- ▶ enhancing technology.
- ▶ Tag needs protection from the adverse weather condition, damage to the tag.

Recycling and Active life Issues

- ▶ Recycling of the tags, environment concern.
- ▶ Active RFID, which consists of battery, has limited life or 2 to 4 years

RFID Technical Challenges

- ▶ Interference: When an organisation uses number of wireless systems, since RFID hotspot also wireless installation, the frequencies may interfere among the systems
- ▶ The systems require effective mitigation from interference

RFID Technical Challenges

- ▶ Effective implementation at data processing subsystem consisting of reader and tag protocols, middleware architecture and EPC standards
- ▶ Needs of low cost of the tags and RFID technology
- ▶ Design robustness
- ▶ Data security

Security Challenges

- ▶ Discovery of foreign attacks (intrusions)
- ▶ Maintain overall data integrity
- ▶ Unauthorised disabling of a tag by a reader which is external, thus making tag useless
- ▶ Unauthorised tag manipulation by a reader which is external, thus making tag useless

Security Challenges

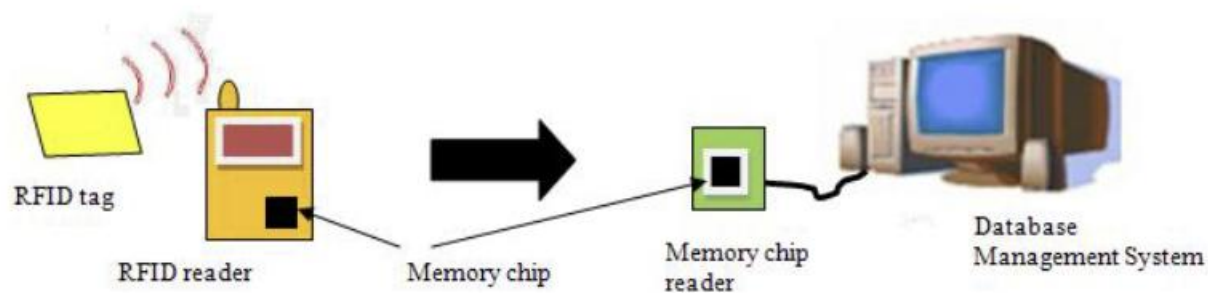
- ▶ Cloning of the tag by unauthorised entity
- ▶ Eavesdropping which means setting up additional reader pretending to reader belonging to the system
- ▶ Man-in-the-Middle attack: When an external object pretends to be either a tag or reader between system tags and readers

Solutions

- ▶ Encryption
- ▶ Tag deactivation on detection of intrusion
- ▶ Mutual authentication between tag and reader,
- ▶ Detection of tag owner
- ▶ Use of read data analyser
- ▶ Data cleaning

Internet of RFIDs

- ▶ Data from the RFID reader after filtering, aggregation and routing store at an IP address
- ▶ Data XML format
- ▶ Data accessing, HTTP and SOAP protocols



IP for an RFID

- ▶ Internet protocol (IP), IPv6 is 128-bit IP address,
- ▶ Needs mapping with 96-bit EPC. (The EPC is header, manufacturer, product and serial number bits.)
- ▶ Secure IPv6 communication, use the cryptographically Generated Addresses (CGAs).

Web of Things of RFIDs

- ▶ WoT means making objects as a part of the World Wide Web.
- ▶ WoT software, architecture styles such as JSON, REST, and programming patterns such as web sockets, makes that feasible
- ▶ WoT data store of objects similar to web pages store.

Summary

We discussed

- ▶ RFID
- ▶ Active and Passive,
- ▶ NFC or UART protocol
- ▶ RFID design issues
- ▶ RFID technology issues
- ▶ RFID security
- ▶ Solutions
- ▶ IPv6 address mapping with EPC
- ▶ Use of CGAs
- ▶ Web of Things of RFIDs

References

- ▶ 1. Book - Internet of Things (Chapter 7)
By Rajkamal

Question



Thank you

